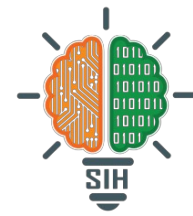


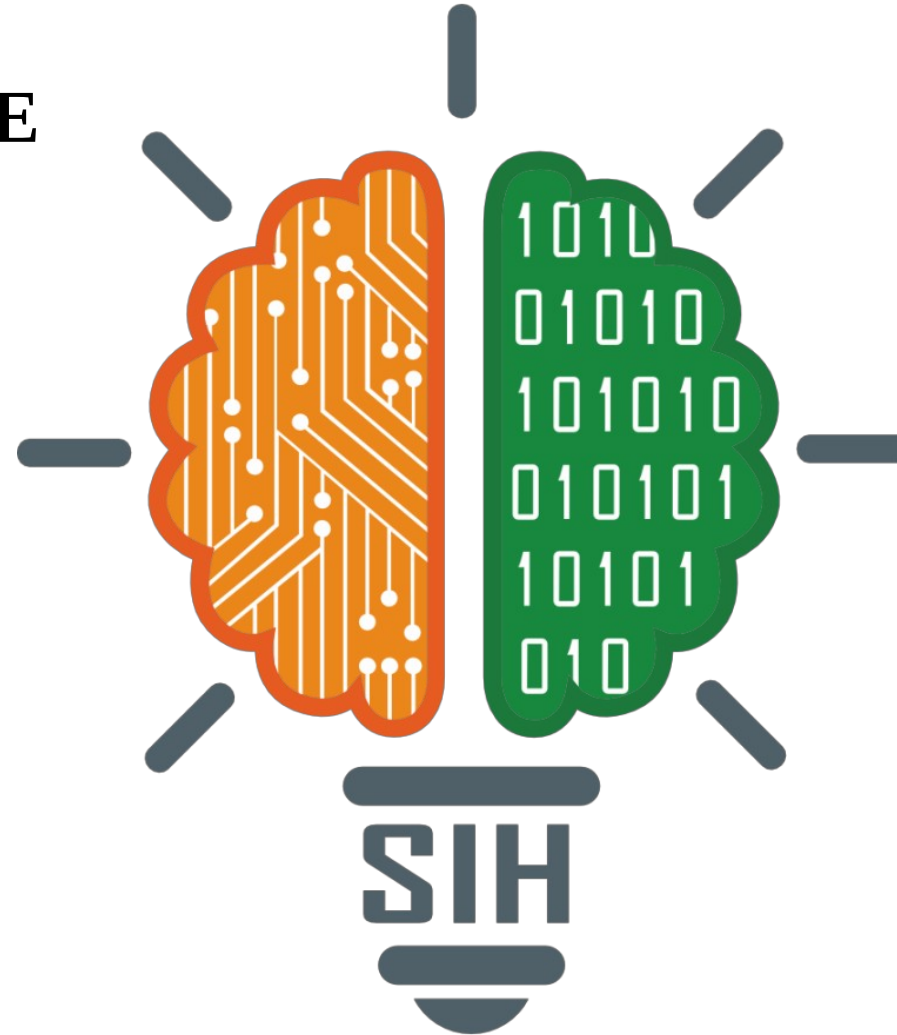
SMART INDIA HACKATHON 2026



SMART INDIA
HACKATHON
2026

TITLE PAGE

- Problem Statement ID – _____
- Problem Statement Title- _____
- Theme- **Artificial Intelligence (AI)**
- PS Category- **Software**
- Team ID- _____
- Team Name (Registered on portal)- **Sankalp**



❖ Proposed Solution (Describe your Idea/Solution/Prototype)

• Detailed explanation of the proposed solution

- Keeps ONE small AI engine in memory – not ten heavy ones
- Snaps on a tiny “skill cap” for the job: coding, maths, writing
- Swaps caps live – no restart, no memory flush, no freeze
- Runs as a silent local server on a loopback address

• How it addresses the problem

Token Tax

→ Everyday tasks run free, on your own PC

GPU Memory Wall

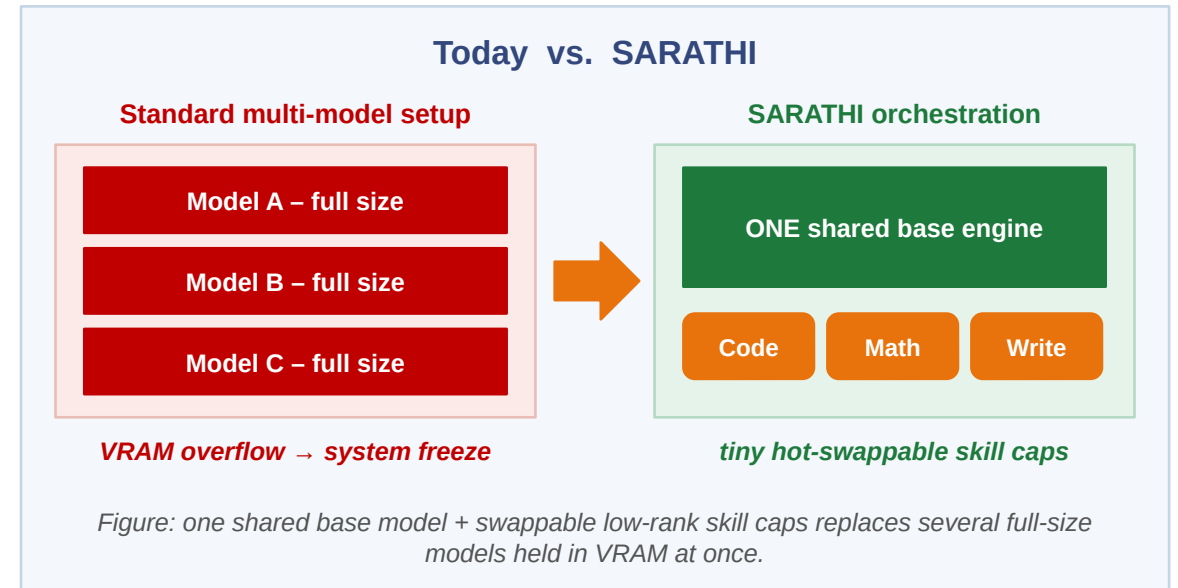
→ One base engine + featherweight skill caps

Context Loss

→ Hybrid memory re-injects history on every swap

• Innovation and uniqueness of the solution

- Auto hardware profiler picks the right-sized free model for YOUR PC
- Live cap-swapping without flushing GPU memory – context survives
- Speaks MCP, so it plugs into the tools you already run

**~78%**

less system VRAM

~99%

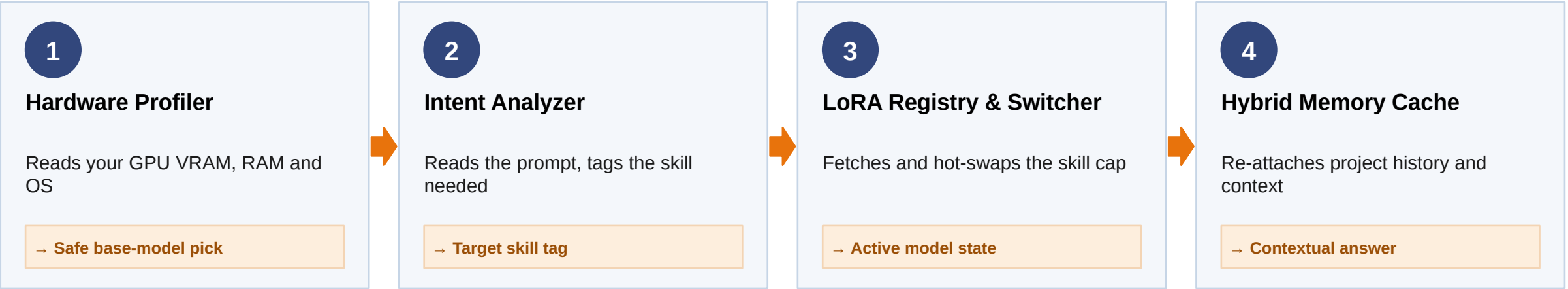
smaller skill files

0 bytes

leave your device

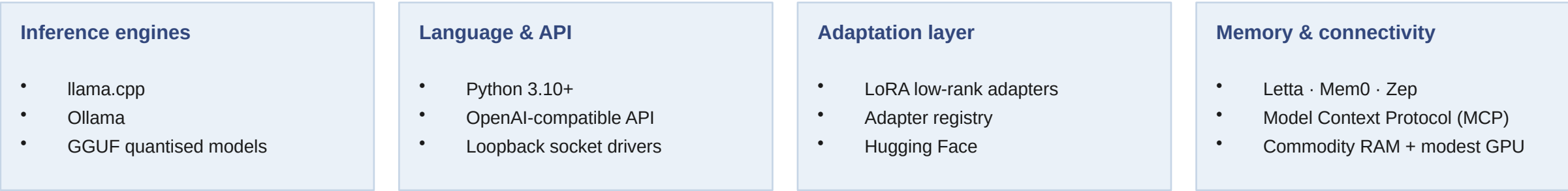
TECHNICAL APPROACH

• Methodology and process for implementation (flow of every prompt)



Runs silently in the background as a local server on a loopback address – nothing leaves the machine during use.

• Technologies to be used (languages, frameworks, hardware)



Adapter fine-tuning is performed externally – SARATHI orchestrates and serves.

• Analysis of the feasibility of the idea

✓

Working proof of concept

A compact base model (Qwen3 4B) already orchestrated with externally fine-tuned adapters.

✓

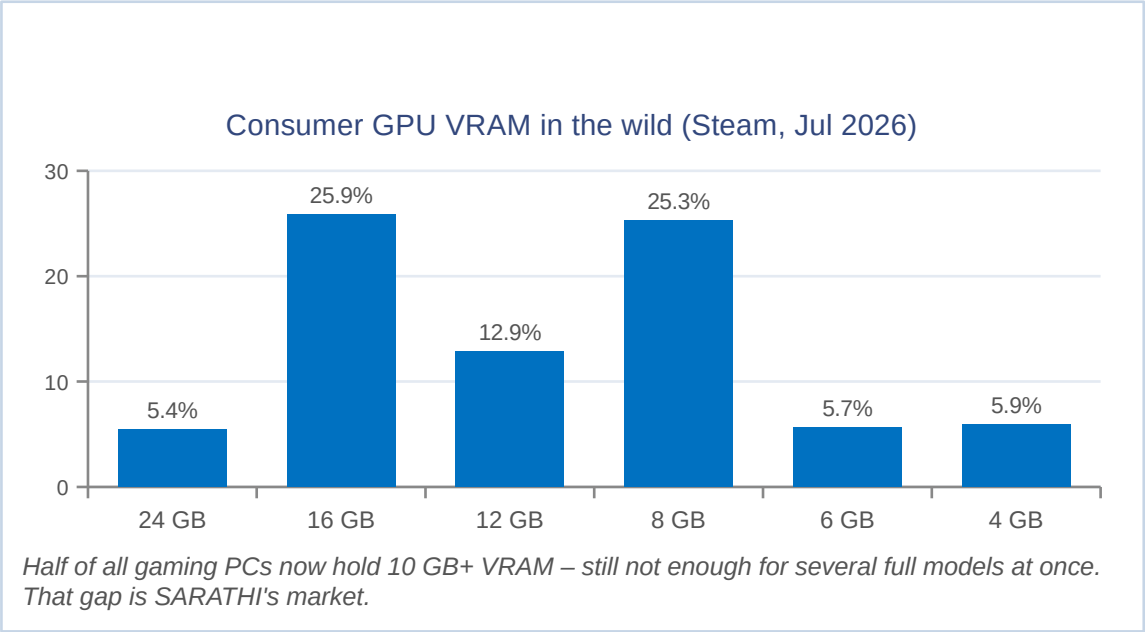
Mature open-source base

llama.cpp and Ollama already support request-level matrix modification.

✓

Modular team split

Profiler, Classifier, Memory Cache and Adapter Registry build and test independently.



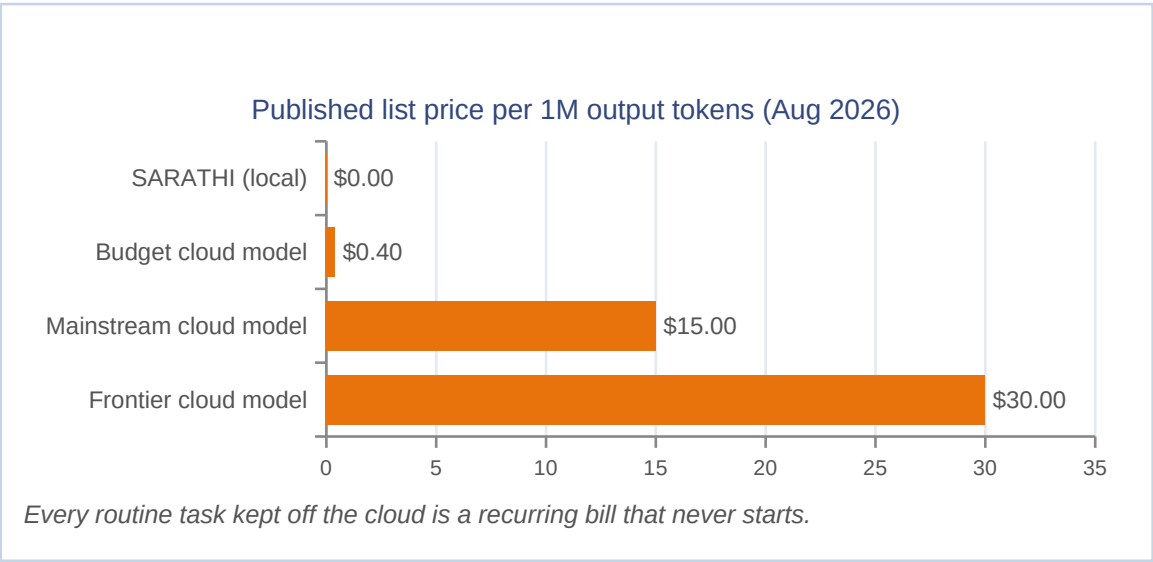
• Potential challenges and risks → Strategies for overcoming these challenges

Risk	Mitigation strategy
Running out of GPU memory	Hardware profiler caps the base model to what the PC can actually hold
A skill cap fails to load	Runtime validation, error boundaries and graceful fallback to the base engine
Unreliable community models	Certification pipeline separates tested base models from experimental ones
Wrong skill picked for a prompt	Fast intent classifier with per-model runtime profiles (chat template, stop tokens)

• Potential impact on the target audience

Students & solo developers Full AI tooling with no monthly bill	Small firms & startups Client code and data never leave the office	Regulated & rural users Works offline; nothing crosses a border
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• Benefits of the solution (social, economic, environmental)



Social	Economic	Environmental
- AI access without a credit card or subscription - Works where the internet does not – offline by design - Personal and medical data stay on the user's device	- Removes per-token fees for everyday tasks - Reuses hardware students and firms already own ~99% smaller skill files – cheap to store and share	- Small local model instead of a data-centre call for simple work ~78% less VRAM → less power drawn per task - Extends the useful life of existing PCs

78% less VRAM used	99% smaller skill files	0 bytes sent to a server	0 rupees per token
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• Details / Links of the reference and research work

1

Hu et al., “LoRA: Low-Rank Adaptation of Large Language Models”, ICLR 2022

Specialising a model needs a tiny fraction of its parameters – proves the “skill cap” is viable.

arxiv.org/abs/2106.09685

2

Chen et al., “Punica: Multi-Tenant LoRA Serving”, arXiv:2310.18547, 2023

Many different skills can be served concurrently over one shared base model.

arxiv.org/abs/2310.18547

3

Sheng et al., “S-LoRA: Serving Thousands of Concurrent LoRA Adapters”, MLSys 2024

Paging adapters just-in-time serves ~2,000 adapters at near-constant throughput; up to 4× vLLM.

arxiv.org/abs/2311.03285

4

Gerganov et al., “llama.cpp: C/C++ LLM Inference Engine”, 2024

Dependency-free engine that makes quantised local execution practical on ordinary PCs.

github.com/ggml-org/llama.cpp

5

Addair et al., “LoRAX: Multi-LoRA Inference Server for Production Workloads”, 2024

Production evidence that dynamic adapter loading works as a served product.

github.com/predibase/lorax

Prototype validated on a compact base model with externally fine-tuned adapters.

Where our numbers come from

- 78% VRAM saving and 99% smaller adapters – SARATHI project synopsis
- GPU VRAM share – Steam Hardware Survey, July 2026
- Token prices – published 2026 LLM API list prices

Standards & ecosystem

- Model Context Protocol – open tool-connection standard, now under the Linux Foundation
- Ollama, Hugging Face, Letta, Mem0, Zep – provider ecosystem

Guided by Prof. Rupali Zambre

Pimpri Chinchwad University, Pune – CSE-AIML